# ชุดเอกสารส่งมอบโปรเจกต์

Tikkie Trip - EV Travel Planner

สำหรับผู้ดูแลและ Claude Code

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Repository: Tikkieteddy/trip-planner
รายงานนี้อ้างอิง source code และเอกสารที่อยู่ใน repository ณ วันที่ระบุ

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# 1. Executive Summary

เว็บแอป Next.js สำหรับวางแผนเดินทางด้วยรถ EV มีการค้นหาสถานทีและสถานีชาร์จ คำนวณเส้นทาง ประเมินแบตเตอรี และจัดเก็บ route แบบ local หรือ Cloud Sync ผ่านบัญชี Auth0

| หัวข้อ | ตรวจพบ |
|---|---|
| Frontend | Next.js App Router, React 19, TypeScript, Tailwind CSS 4 |
| Hosting | Vercel project trip-planner; production settings NOT VERIFIED |
| Identity | Auth0 Next.js SDK; cloud handler scopes by session.user.sub |
| Persistence | Upstash Redis REST; no relational schema or migrations found |
| Map APIs | Google Maps JavaScript, Places API (New), Routes API |

สถานะ: source implementation พบแล้วหลายส่วน แต่ login/Redis/Google live flows, physical mobile QA และ current deployment configuration ยังต้องตรวจแยกต่างหาก

# 2. Architecture and Data Flow

Browser ส่ง JSON ไป Next.js Route Handlers; handlers validate ด้วย Zod และเรียก Google Places/Routes ด้วย server key. Maps JavaScript API โหลดใน browser ด้วย public restricted key. Auth0 session ปกป้อง cloud route API; Upstash Redis จัดเก็บ route ต่อ subject และ config โฆษณาแบบ global.

| From | To | Purpose |
|---|---|---|
| Browser | Next.js APIs | Place search, route request, cloud route actions |
| Next.js | Google Places | Autocomplete, text/nearby/charger search and details |
| Next.js | Google Routes | Compute driving route and legs |
| Next.js | Upstash Redis | Cloud route library and Ads configuration |
| Browser | Maps JS | Render map, route geometry and markers |

# 3. Screens and Navigation

พบ 2 หน้าหลัก: / คือ Trip Planner; /cms คือ Ads CMS. /admin/ads redirect ไป /cms. Tutorial 27 ขัน, tabs, menus และ dialogs เป็น states ในหน้า ไม่ใช่ routes แยก. Wireframes ใน dashboard เป็น schematic จาก components ไม่ใช่ภาพ pixel-perfect.

- Planner: trip settings, vehicle profile, filters, map search, route panel, itinerary, chargers, nearby places, account/saved routes.
- CMS: admin password login, mobile/desktop script fields, enabled toggle, save/logout and setup/storage error states.

# 4. Feature Inventory

Implemented in source: trip setup, map and place search, Google route handler, charger along route and polyline segmentation, battery estimate, nearby categories, route serialization, tutorial and shared credit footer. Partial/service-dependent: Auth0, Redis, EV Station PluZ data coverage, app links, touch/phone behavior and Ads. NOT FOUND in repo scan: unit/e2e test suite, SQL schema/migrations and GitHub workflow files.

Audit count: 20 รายการ — Implemented 7, Partial 9, Deprecated 1 และ NOT FOUND 3. Implemented ยืนยันเฉพาะว่าพบโค้ด; ไม่ได้แปลว่า production integration ผ่านการตรวจแล้ว.

| Feature group | Current status | Evidence / next verification |
|---|---|---|
| Planner + route | Implemented in source | Run one-way and round-trip end-to-end |
| Charging suggestions | Implemented / provider-dependent | Verify real results, long routes, coverage |
| Cloud route sync | Partial | Auth0 + Redis test, isolation and second device |
| Ads CMS | Partial; last priority | Provider rules, consent, CSP, storage and login |
| Automated testing | NOT FOUND | npm test currently aliases build |

# 5. API and Storage

| Method / path | Access | Summary |
|---|---|---|
| POST /api/places | Public | Autocomplete, text, route charger and nearby search |
| POST /api/place-details | Public | Place ID to normalized details |
| POST /api/routes | Public | Driving route, legs, polyline, modifiers |
| GET/PUT /api/cloud-routes | Auth0 | Read/write max 25 routes per user |
| GET /api/ads | Public | Only enabled scripts; no-store |
| POST /api/admin/login/logout | Admin cookie | HMAC session exchange / clear |
| GET/PUT /api/admin/ads | Admin cookie | Read/write enabled state and scripts |

Redis key families: trip-planner:cloud-routes:v1: stores routes + updatedAt; trip-planner:ads:primary stores enabled, mobileScript, desktopScript and updatedAt. No SQL entities were found.

# 6. Setup and Verification Commands

npm ci npm run dev npm run lint npm run build npm test # currently runs build only

Environment names found include NEXT_PUBLIC_GOOGLE_MAPS_BROWSER_KEY, NEXT_PUBLIC_GOOGLE_MAPS_MAP_ID, GOOGLE_MAPS_SERVER_KEY, Auth0 SDK settings, UPSTASH_REDIS_REST_URL/TOKEN, ADS_ADMIN_PASSWORD, ADS_ADMIN_SESSION_SECRET and optional Clerk migration secret. Values are intentionally omitted. Verify exact Auth0 SDK variables from installed SDK docs and dashboard.

# 7. Risks and Limitations

- Google APIs can incur billing; results depend on data coverage, fields, restrictions and quota.
- Never expose server key, Redis token, Auth0 secret or admin secrets; .env.local is excluded.
- Cloud route data is keyed by Auth0 subject; legacy Clerk migration is not verified on real accounts.
- Ads scripts execute in the browser and require trusted admin, provider policy, consent/privacy and CSP review.
- Battery/range is an estimate based on user inputs; it is not a vehicle telemetry guarantee.
- No unit/e2e tests found; `npm test` is only a build alias.

# 8. Priority Roadmap

- P0: verify Auth0 sign-in, Redis route save/read, user isolation, cross-device sync and redirect recovery.
- P1: exercise real one-way/round-trip Routes + Places/chargers with a controlled small trip and observe quota.
- P1: add automated tests for schemas, polyline segmentation, battery edge cases and auth/storage failures.
- P2: physical phone QA for map controls, tutorial, result scrolling, share links and footer tooltip.
- P3, last: review and test Ads provider, consent/security and CMS before enabling monetization.

# 9. Claude Code Takeover Checklist

- Read CLAUDE.md, HANDOVER.md, DOCUMENT_INDEX.md, DEVELOPMENT_ROADMAP.md and NEXT_STEPS.md.
- Inspect git status and verify source before trusting any document.
- Validate environment by presence only; never print secret values.
- Run lint/build; separate source evidence from external production verification.
- Do not deploy, push or alter production data without explicit authorization.

# 10. Validation Boundaries

This handover records repository evidence only. Production login, Vercel configuration, Redis availability, Google quota/results, real-user migration, physical-device screenshots and current deployment were not tested as part of document authoring. See HANDOVER_VALIDATION.md for the checklist and per-item status.

Source of truth: repository source. When it conflicts with this report, inspect the source and update the handover.